

This is a comprehensive, silicon-validation-grade test suite for **NVMe 2.0d Boot Partitions**. As an architect, I have categorized these into **six functional domains** to ensure 100% coverage of the specification, including the state machine transitions, register boundary conditions, and the **TP 4170** security lifecycle.

NVMe 2.0d Boot Partition Master Test Matrix

Domain 1: Capability & Register Discovery (Basic Compliance)

Test ID	Test Name	Objective	Steps	Expectations
BP_01	CAP Register Check	Verify BPS bit.	Read CAP (Offset 00h).	Bit 15 (BPS) must be 1.
BP_02	BPINFO Default State	Verify reset values.	Read BPINFO (Offset 40h) after POR.	BPSZ > 0, BRS = 00b, ABPID = 0 or 1.
BP_03	Register Read/Write	Verify R/W permissions.	Attempt write to BPINFO; Read/Write to BPSEL.	BPINFO is Read-Only. BPSEL/BPRMBL are R/W.
BP_04	Alignment Constraint	Test 4KB alignment.	Write non-4KB aligned addr to BPRMBL.	Controller should ignore lower 12 bits or error out.

Domain 2: Pre-Enable Boot Operations (The "PBL" Flow)

These tests must be performed while CC.EN = 0.

Test ID	Test Name	Objective	Steps	Expectations
BP_05	Pre-Enable Read BPO	Load SBL from BPO.	Trigger BPSEL (BPID=0) while CC.EN=0.	BRS transitions 01b -> 10b.

BP_06	Pre-Enable Read BP1	Load SBL from BP1.	Trigger BPSEL (BPID=1) while CC.EN=0.	Successful DMA from BP1.
BP_07	Consecutive Reads	Verify state machine reset.	Trigger read BPO; wait for Success; trigger BP1.	Both must succeed consecutively.
BP_08	BRS Polling	Verify BRS timing.	High-frequency poll of BRS during DMA.	Must see 01b (In Progress) before 10b.
BP_09	DMA Interruption	Test Reset during DMA.	Trigger Read; Toggle CSTS.RDY or PCI Reset.	BRS must return to 00b immediately.

Domain 3: Redundancy & Active Partition Management

Test ID	Test Name	Objective	Steps	Expectations
BP_10	ABPID Identification	Verify active pointer.	Read BPINFO.ABPID.	Must match the image currently used for boot.
BP_11	Commit Action 7 (Switch)	Verify partition swap.	Firmware Commit (Action 111b) to BPID 1.	BPINFO.ABPID changes to 1.
BP_12	Active/Inactive Logic	Verify read of inactive.	Read from BPID opposite of ABPID.	Controller must allow reading inactive BP.
BP_13	Persistent	Verify NVM	Swap ABPID; Power Cycle;	New ABPID must persist

	ABPID	storage.	Read ABPID.	across power loss.
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Domain 4: Image Update & Admin Commands

Test ID	Test Name	Objective	Steps	Expectations
BP_14	FW Image Download	Test data buffering.	Use Admin Command (01h) to send BP image.	Command completes with Success.
BP_15	FW Commit Action 6	Update Inactive BP.	Commit downloaded image to inactive BP.	BP updated; ABPID does NOT change.
BP_16	FW Commit Action 7	Update & Swap.	Commit downloaded image to inactive + Swap.	BP updated AND ABPID flips.
BP_17	Partial Update Error	Test size mismatch.	Commit image smaller than partition size.	Controller must zero-fill or reject.
BP_18	Large Image Rejection	Test overflow update.	Commit image > BPSZ.	Command fails: "Invalid Field in Command".

Domain 5: TP 4170 Security & Write Protection

Test ID	Test Name	Objective	Steps	Expectations
BP_19	Get Log Page	Verify Log	Fetch Log ID	Reports BP count, size,

	15h	Support.	15h.	and lock status.
BP_20	Set Features (Lock BP)	Lock BPO until Reset.	Set Feature 15h (BPWP=01b).	Log 15h shows BPO is "Locked".
BP_21	Locked Write Rejection	Verify protection.	Attempt FW Commit to a Locked BP.	Status Code: Access Denied (86h) .
BP_22	Permanent Lock	Test "Write Once" logic.	Set Feature 15h (BPWP=10b).	BP is locked; lock persists after Power Cycle.
BP_23	Read While Locked	Verify read access.	Trigger BPSEL read on a Locked BP.	Read must still succeed (Lock only affects write).
BP_24	Unlock Sequence	Verify secure transition.	Issue unlock (if auth supported).	Log 15h Bit 1 clears to '0'.

Domain 6: Negative Testing & Boundary Conditions

Test ID	Test Name	Objective	Steps	Expectations
BP_25	Offset Overflow	Read beyond BPSZ.	Set BPOFF > BPSZ. Trigger read.	BRS = 11b (Error).
BP_26	Size Wrapping	Read spanning boundary.	Set BPOFF + BPSIZE > BPSZ.	BRS = 11b (Error).
BP_27	Invalid BPID	Select non-existent	Set BPID = 2 in BPSEL.	Controller must flag Error

		BP.		in BRS.
BP_28	Unaligned Size	Test 4KB granularity.	Request 2KB via BPSIZE (if not supported).	Controller must round up or Error.
BP_29	Concurrent Read/EN	Trigger during CC.EN=1.	Trigger BPSEL read while I/O queues active.	Should succeed (Runtime read supported).
BP_30	Multiple Triggers	Back-to-back BPSEL.	Write BPSEL twice before 1st DMA finishes.	Controller must ignore 2nd or restart.
BP_31	Log Page 15h vs Info	Data consistency.	Compare Log 15h BPSZ vs BPINFO.BPSZ.	Must be mathematically identical.
BP_32	Hot Reset Recovery	Stress test resets.	Trigger 1000 PCI Resets during BP reads.	No controller hangs; BRS always recovers.

Execution Guide for Testers

1. **Hardware setup:** Use a PCIe protocol analyzer (like Teledyne LeCroy) to verify that the DMA bursts for the Boot Partition do not use NVMe Admin/IO Queues.
2. **Compliance Warning:** Many consumer SSDs do not support **TP 4170**. If **Log Page 15h** returns "Invalid Field," the device is not compliant with the latest security revisions of NVMe 2.0d.
3. **Observation Column:** Always document the **PCIe Link Speed** during the transfer. Some controllers down-train to Gen1 during the pre-enable phase, which can impact boot timing.